

ALEXANDER DORNBACK

(412) 925-1074 | alexander.dornback@gmail.com | [LinkedIn](#) | [Portfolio](#)
MS, Robotics @ Columbia | Controls, Hardware, Sim2Real | Available August 2026

EDUCATION

Columbia University MS, Mechanical Engineering , 3.95/4 Robotics, Automation, and Controls Concentration <i>Research Affiliation: Creative Machines Laboratory, Prof. Hod Lipson (January 2026 – Present)</i>	New York, NY Expected Dec 2026
Case Western Reserve University BS, Biomedical Engineering Biomechanics Concentration Minor, Music	Cleveland, OH May 2022

ENGINEERING PROJECTS

FIDO: an open-source, trainable, 3D-printed quadruped robot

- Implemented random search/hill-climbing for gait parameter optimization and PPO for sensorless joint neutral estimation
- Achieved stance in 10 minutes of parallel-environment PPO training on 4 custom linkage-drive legs in MuJoCo
- Achieved stable walking in 30-minute training and completed sim-to-real pipeline for gait of 25 cm/s on real-world deployment

CATCH: Hybrid MPC-CBF impact shaping for bipedal locomotion

- Selected CBF after PID and vanilla MPC exceeded 3.34N actuator limits on impact; enforced safety across drops up to 40cm
- Built kinematic TTC vision estimator as feedforward input to dynamically adjust prediction horizon pre-impact
- At 40cm, MPC-CBF achieved 6.71N GRF vs. 8.00N for PID – a 16% reduction while ensuring hardware limit adherence

TYPHOON: A novel polar IDEX system for simultaneous multi-material 3D printing

- Designed and validated a polar dual-hotend 3D printer with simultaneous multi-material deposition and zero-purge waste
- Simulink dynamic analysis for polar axis stepper selection revealed 2-2.5x resonance amplification missed in static motor sizing

WORK EXPERIENCE

Serpent Robotics Mechatronics Intern	Philadelphia, PA June 2026 – Present
• Designing vertical cable-driven parallel robot with 2 capstan winches and rope ascender for autonomous in-tree positioning • Developing active tensioner architecture with PID control in embedded C++ (ESP32) on rope tension via Hall-effect dancer • Performing FMEA and Python parametric modeling on worm-gear automatic saw arm to inform motor selections • Contributing to ROS 2 migration and MuJoCo CDPR digital twin ahead of July pilots with utility companies and arborists	
Weill Cornell Medicine Dalio Institute Robotics Engineer	New York, NY Jul 2022 - Sep 2025
• Led 3 engineers over 12 months to deliver a clinically validated intubation robot tested by untrained and ENT/EMS operators • Designed a superelastic nitinol tendon-driven actuator with scotch-yoke transmission, achieving 30°/s tip deflection • Embedded C++ (Arduino) for IR closed-loop control achieved first-pass intubation in 20-second benchmark with 90% accuracy • Designed and fabricated 3 mechanical testbeds tracking twist, stretch, and bend of a silicone capacitive force sensor • Built CFD-validated multi-material segmented printing workflow for 65 CT angiograms to assess coronary artery disease risk	
Haima Therapeutics Formulation Scientist	Cleveland, OH Oct 2020 - Jun 2022
• Characterized over 200 nanoparticle batches, raising \$3m in DARPA grant funding alongside C-suite and scientist team • Co-author: Sekhon et al. Wiley Adv. Science, 2025 (https://doi.org/10.1002/adv.202600002)	

SKILLS

- Controls: Model Predictive Control, Control Barrier Functions, PID, RL (PPO), Sim2Real
- Software: Python, ROS 2, C++, MATLAB/Simulink, I2C/SPI, Linux, Git
- Mechatronics: Embedded C++ (ESP32/Arduino/Pi Zero), Sensor Fusion, Actuator/Motor Control, Transmission Design
- Modeling and Design: MuJoCo, Gazebo, Kinematics (FK/IK), State Estimation, Forward Modeling, SolidWorks (CSWA, Sim)